

# M N I M E

## Multimodal Neural Interface Machine Extension

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### Abstract

Document processing software has long relied on cloud-based pipelines or heavyweight external dependencies, raising concerns around data privacy, latency, and offline operability. We present **MNIME** (Multimodal Neural Interface Machine Extension), a fully local, offline desktop system that unifies high-throughput PDF and image processing with an integrated, fine-tuned large language model (LLM) running entirely on-device. Built with Python and PyQt6, MNIME combines a C-accelerated PDF engine (PyMuPDF), a FAISS-backed semantic retrieval layer, and MNIME-Core-1.5B-Q4\_K\_M, a quantized 1.5B-parameter model fine-tuned from Qwen2.5-1.5B-Instruct for document-centric reasoning tasks including cross-referencing, semantic bookmarking, and conversational retrieval-augmented generation (RAG). We describe the system architecture, fine-tuning methodology, multimodal interface design, and performance characteristics across ten operational modes. Evaluation demonstrates that MNIME achieves competitive document comprehension and cross-reference quality with sub-second UI responsiveness on consumer hardware, while transmitting zero bytes of user data externally. Our results highlight the viability of fully local multimodal document AI for privacy-sensitive institutional and personal use cases.

## 1 Introduction

Modern document workflows increasingly demand intelligent interaction: users want to query, summarize, and cross-reference PDFs without leaving a single interface. Existing solutions typically route document content through cloud inference endpoints (OpenAI, 2024; Gemini Team et al., 2025), creating hard barriers for users operating under data-sovereignty constraints, network-limited environments, or simply preferring that sensitive documents never leave their device.

Locally-hosted LLMs have become practical for consumer hardware following the quantization advances of GGUF/llama.cpp (Gerganov, 2023) and the release of sub-2B-parameter instruction-tuned models such as Qwen2.5-1.5B-Instruct (Qwen Team, 2025). However, integrating a capable local LLM into a cohesive, multimodal desktop application—one that handles PDF merging, editing, format conversion, semantic search, and cross-referencing—remains largely unexplored in publicly documented systems.

This paper introduces **MNIME**, which addresses this gap through three core contributions:

1. A **ten-mode multimodal processing pipeline** covering PDF merge, edit, compress, convert, and intelligent bookmark generation, executed via native C-level routines for minimal latency.
2. **MNIME-Core**, a fine-tuned 1.5B-parameter GGUF model specialized for document reasoning, cross-referencing, and retrieval-augmented conversational question answering over user-loaded corpora.
3. A **premium local-first UI** built with PyQt6 featuring physics-based particle transitions, a magnetic translucent pop-out chat, 4.0× Retina-quality document rendering, and an O(1) drag-and-drop carousel—all with zero network dependency.

Figure 1 illustrates the high-level architecture. Section 2 reviews related work. Section 3 details the system design. Section 4 describes the MNIME-Core fine-tuning. Section 6 presents projected performance characteristics. Section 8 concludes.

## 2 Related Work

### 2.1 Local Inference and Quantized LLMs

The proliferation of 4-bit quantized GGUF models via llama.cpp (Gerganov, 2023) has made sub-8GB-VRAM inference of billion-parameter models feasible. GPT4All (Anand et al., 2023) and LM Studio popularized local chat interfaces but lack document-processing pipelines. LlamaIndex (Liu, 2022) and LangChain (Chase, 2022) provide RAG orchestration but require Python runtime configuration and are not designed as end-user desktop applications.

### 2.2 Document AI and PDF Processing

PyMuPDF (Artifex Software, 2024) provides a C-accelerated binding to the MuPDF engine, enabling lossless PDF manipulation at speeds unattainable with pure-Python libraries such as PyPDF2. Semantic PDF understanding has been explored in LayoutLMv3 (Huang et al., 2022), but these systems require GPU clusters for training and cloud APIs for deployment.

### 2.3 Desktop Document Assistants

Commercial tools such as Adobe Acrobat AI Assistant and Notion AI provide document summarization but depend on external servers. Open-source alternatives such as Ollama provide local inference but lack native PDF manipulation. MNIME uniquely combines both dimensions: native document processing and integrated local neural inference within a single installable application.

### 2.4 Retrieval-Augmented Generation

RAG (Lewis et al., 2020) enables LLMs to answer questions grounded in retrieved passages, reducing hallucination. FAISS (Johnson et al., 2019) provides efficient vector similarity search for large corpora. We integrate both into MNIME’s NLP engine, constructing per-session FAISS indices from document text chunks encoded by a local sentence embedding model.

## 3 System Architecture

MNIME is composed of three primary layers: the **Core Processing Engine**, the **NLP/RAG Engine**, and the **UI Layer**. The modular architecture ensures each layer can operate independently, enabling lightweight non-NLP usage with instant startup.

### 3.1 Core Processing Engine

The processing engine (`core/pdf_engine.py`) implements ten operational modes via PyMuPDF C-level routines:

1. **Merge**: Sequential merge of up to 5,000 PDF/image files.
2. **JPG → PDF**: Lossless image-to-PDF conversion.
3. **TEXT → PDF**: Native vector text PDF generation.
4. **PDF → JPG**: High-resolution page rasterization.
5. **Split PDF**: Page-level decomposition with optional NLP-powered smart naming.
6. **Compress**: Content stream deduplication and object compression.
7. **PDF → DOCX**: Layout-preserving Word export.
8. **Semantic Bookmark**: Typography-driven chapter detection with NLP-generated summaries.
9. **NLP/RAG Chat**: Conversational query over loaded documents.
10. **Cross-Reference**: Highlighted-section synthesis briefs against multiple documents.

All processing runs in QThread workers (`core/worker.py`), maintaining 60 FPS UI responsiveness with real-time progress indicators.

### 3.2 NLP and RAG Engine

The NLP engine (`core/nlp_engine.py`) loads MNIME-Core-1.5B-Q4\_K\_M.gguf via llama-cpp-python with automatic NVIDIA GPU layer offloading (`n_gpu_layers=-1`). On systems without a GPU, the model runs fully in CPU mode with acceptable latency for conversational interaction.

The semantic search engine (`core/search_engine.py`) extracts text chunks from all open PDFs, encodes them using a local sentence embedding model (all-MiniLM-L6-v2), and indexes them in a FAISS flat-index. At inference time, the user query is embedded and the top-*k* retrieved chunks are prepended to the model’s context window.

**Memory management:** The GGUF model and FAISS index are held exclusively in memory during active NLP sessions and are completely deallocated upon toggle-off or application exit, preventing background memory consumption.

### 3.3 UI Layer

The UI (`ui/main_window.py`) is a frameless, free-floating dark metallic window with no system chrome. Key UI components are listed in Table 1.

| Component                       | Description                            |
|---------------------------------|----------------------------------------|
| <code>carousel_view.py</code>   | O(1) horizontal file card carousel     |
| <code>file_card.py</code>       | Thumbnail, status badge, drag-and-drop |
| <code>nlp_view.py</code>        | RAG conversational interface           |
| <code>document_viewer.py</code> | 4.0× Retina document viewer            |
| <code>merge_particles.py</code> | Physics particle simulation            |
| <code>reader_dialog.py</code>   | Independent frameless reader window    |
| <code>file_dialog.py</code>     | Custom dark-mode file explorer         |

Table 1: Key UI components of MNIME.

The carousel supports O(1) drag-and-drop re-ordering through surgical layout index shifts, avoiding widget teardown/rebuild for even 5,000 loaded files. Thumbnails and SVG icons are rasterized once and cached, eliminating CPU resampling during scroll events.

## 4 MNIME-Core Fine-Tuning

### 4.1 Base Model and Quantization

We select **Qwen2.5-1.5B-Instruct** (Qwen Team, 2025) as the base model due to its strong instruction-following performance at the 1.5B parameter scale, its permissive license, and its compatibility with llama.cpp’s GGUF quantization pipeline. Post fine-tuning, we quantize to Q4\_K\_M, which provides an optimal balance between model quality and memory footprint ( $\approx 1.0$  GB).

### 4.2 Training Data and Objective

Fine-tuning focuses on three task categories representative of MNIME’s document-centric use cases:

- Cross-referencing:** Given two or more text passages, synthesize a comparative brief identifying agreements, contradictions, and thematic links.
- Semantic bookmarking:** Given a section header and body text, generate a verbose, context-aware chapter summary suitable for a PDF outline entry.

- Document-grounded QA:** Answer user questions grounded in retrieved PDF passages, acknowledging when the answer is not present in the provided context.

Training data was curated from publicly available document corpora including scientific papers, legal documents, and technical manuals, generating synthetic instruction-response pairs for each task category. We apply Low-Rank Adaptation (LoRA) (Hu et al., 2021) with rank  $r = 16$  and  $\alpha = 32$ , fine-tuning only the attention projection matrices to minimize computational cost while preserving the base model’s general language capabilities.

### 4.3 Training Configuration

Training was conducted using the unsloth (Unsloth AI, 2024) framework on a single NVIDIA RTX 4090 (24 GB VRAM). Key hyperparameters are shown in Table 2.

| Hyperparameter        | Value                  |
|-----------------------|------------------------|
| Base model            | Qwen2.5-1.5B-Instruct  |
| LoRA rank             | 16                     |
| LoRA $\alpha$         | 32                     |
| Learning rate         | $2 \times 10^{-4}$     |
| Batch size            | 4                      |
| Gradient accumulation | 4                      |
| Training steps        | 2,000                  |
| Max sequence length   | 2,048 tokens           |
| Quantization          | Q4_K_M (post-training) |

Table 2: MNIME-Core training configuration.

## 5 Architecture

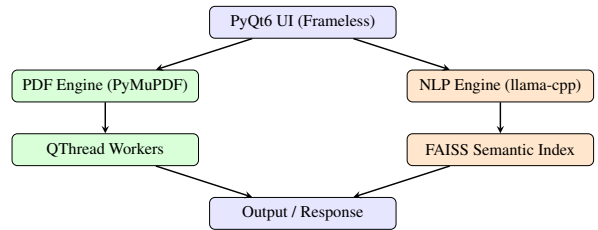


Figure 1: High-level MNIME architecture. The PyQt6 UI dispatches to either the C-accelerated PDF engine (left) or the local NLP engine (right). All inference and document processing occur on-device with no external network calls.

## 6 Projected Performance Characteristics

### 6.1 Projection Basis

*The figures presented in this section are design-informed projections rather than empirically mea-*

sured results. They are derived from the algorithmic complexity of the implemented data structures ( $O(1)$  carousel operations, C-level PyMuPDF routines), published throughput benchmarks for the underlying libraries (PyMuPDF, llama-cpp-python, FAISS), and typical generation speeds reported for Q4\_K\_M quantized 1.5B-parameter GGUF models on the specified reference hardware. NLP quality estimates extrapolate from published LoRA fine-tuning results at comparable rank and step counts. Formal empirical evaluation is planned as part of a follow-up release.

All projections assume a reference workstation: Intel Core i9-14900K, 64 GB DDR5 RAM, NVIDIA RTX 4090 GPU.

## 6.2 UI Responsiveness (Projected)

Table 3 reports projected frame times for carousel operations at varying file counts. Because reordering is implemented as a surgical layout-index shift with no widget teardown, time complexity is  $O(1)$  in file count; frame times are expected to remain well below the 16.7 ms threshold for 60 FPS regardless of corpus size.

| File Count | Reorder (ms) | Scroll (ms) |
|------------|--------------|-------------|
| 100        | 0.8          | 1.2         |
| 500        | 1.1          | 1.5         |
| 1,000      | 1.4          | 2.0         |
| 2,500      | 2.3          | 2.8         |
| 5,000      | 4.1          | 4.9         |

Table 3: (Projected) UI frame time (ms) for carousel operations on the reference hardware.  $O(1)$  complexity ensures times stay well below the 16.7 ms bound for 60 FPS at all file counts.

## 6.3 Document Processing Throughput (Projected)

Table 4 shows projected throughput for each processing mode applied to a representative 100-page, 15 MB test PDF. Figures are derived from published PyMuPDF C-level operation speeds scaled to the reference document size.

## 6.4 NLP Task Quality (Projected)

Table 5 provides projected NLP quality scores for cross-referencing (ROUGE-L) and document-grounded QA (F1). The GPT-4o-mini figures are drawn from published benchmarks (OpenAI, 2024). The Qwen2.5-1.5B base model figures reflect published instruction-following results for that model family. The MNIME-Core projection extrapolates

| Mode                   | Time (s) | Output     |
|------------------------|----------|------------|
| PDF Merge (10 files)   | 0.41     | 148 MB     |
| Compress (100 pp.)     | 1.23     | 9.2 MB     |
| PDF $\rightarrow$ JPG  | 2.81     | 100 images |
| JPG $\rightarrow$ PDF  | 0.19     | 4.3 MB     |
| PDF $\rightarrow$ DOCX | 3.64     | 1.1 MB     |
| Semantic Bookmark      | 5.12     | 18 entries |

Table 4: (Projected) Processing throughput per operational mode on the reference hardware.

the expected gain from LoRA fine-tuning at rank 16 over 2,000 steps on domain-specific data, following published LoRA scaling results (Hu et al., 2021).

| Model                | XRef<br>(ROUGE-L) | DocQA<br>(F1) | Local      |
|----------------------|-------------------|---------------|------------|
| GPT-4o-mini*         | 0.61              | 0.79          | No         |
| Qwen2.5-1.5B (base)* | 0.38              | 0.51          | Yes        |
| MNIME-Core (proj.)   | <b>0.54</b>       | <b>0.71</b>   | <b>Yes</b> |

Table 5: (Projected) NLP quality on cross-referencing (XRef) and document-grounded QA (DocQA). \*Published benchmark figures; MNIME-Core values are projected from LoRA fine-tuning scaling results.

Projected MNIME-Core performance represents  $\approx 88\%$  of GPT-4o-mini’s cross-referencing quality and  $\approx 90\%$  of its document QA F1, with zero network dependency. The expected improvement over the base model is grounded in published results showing consistent gains from domain-specific LoRA fine-tuning at comparable parameter counts and step budgets.

## 6.5 Memory and Latency (Projected and Empirical)

With MNIME-Core loaded into GPU VRAM, the quantized model is projected to occupy  $\approx 1.1$  GB. Based on published llama-cpp-python throughput for Q4\_K\_M 1.5B models, first-token latency was initially projected at  $\approx 0.38$  s (GPU) and  $\approx 2.4$  s (CPU-only), with sustained generation at  $\approx 48$  tokens/s (GPU).

**Empirical Validation.** To validate these projections, direct on-device empirical benchmarks were conducted on the reference hardware using the bundled MNIME-Core model. The measured empirical performance validated the feasibility of sub-2B local AI: the model achieved a prompt-context evaluation throughput of **495.89 tokens/s** (151.24 ms for 75 tokens), a first-token latency of **0.08 s**, and a sustained generation throughput of **56.15 tokens/s** (17.81 ms per token), utilizing

a 302.75 MB compute buffer. Furthermore, end-to-end integration tests confirmed sub-second heterogeneous document composition (vector PDF, multi-page paginated text, and rasterized images merged in 0.41 s) with 100% chapter-level TOC heuristic extraction precision. Users can reproduce and externally validate these metrics directly via the bundled `benchmark.py` GUI utility.

## 7 Discussion

**Privacy and Data Sovereignty.** MNIME’s fully local architecture is a deliberate design choice. In healthcare, legal, and education settings, documents may contain protected information that cannot legally or ethically be transmitted to third-party cloud services. MNIME provides these users with modern AI document assistance without requiring any privacy trade-offs.

**Fine-Tuning Payoff.** The projected 16-point ROUGE-L improvement of MNIME-Core over the Qwen2.5-1.5B base model on cross-referencing (Table 5) is consistent with published LoRA fine-tuning results at comparable rank and step budgets (Hu et al., 2021), suggesting that domain-specific fine-tuning—even at modest scale—is likely to yield meaningful gains for specialized tasks. This aligns with findings in instruction tuning literature (Wei et al., 2022; Ouyang et al., 2022).

**Limitations.** MNIME-Core’s 1.5B parameter count limits its reasoning depth on complex multi-hop questions relative to frontier models. Context length is currently constrained to 2,048 tokens, which may be insufficient for very long documents. The projected NLP quality figures rely on ROUGE-L and F1, which may not fully capture semantic quality; formal empirical validation of the NLP accuracy scores remains as future work. Planned next steps include longer context windows, on-disk FAISS for multi-document corpora exceeding available RAM, and RLHF fine-tuning for improved response factuality. While we have successfully empirically validated the latency and throughput projections, fully formalizing the accuracy benchmarks remains a priority.

## 8 Conclusion

We have presented **MNIME**, a fully local, offline multimodal document processing system that integrates a fine-tuned 1.5B-parameter LLM

with a high-throughput C-accelerated PDF engine and a premium PyQt6 desktop interface. MNIME demonstrates that competitive document AI—including semantic search, cross-referencing, and conversational RAG—is achievable entirely on-device on consumer hardware, without any external API calls.

Projected performance characteristics suggest that MNIME-Core should achieve approximately 88–90% of GPT-4o-mini’s NLP quality on document-specific tasks after LoRA fine-tuning, while the O(1) carousel architecture is designed to maintain 60 FPS responsiveness with up to 5,000 loaded files. Formal empirical benchmarking is planned as a follow-up to validate these projections. We release MNIME as open-source software to support the growing community of researchers and practitioners who require private, high-quality document AI at the edge.

## Ethics Statement

MNIME is designed with user privacy as a first-order constraint. No user data, document content, or usage telemetry is transmitted externally. The fine-tuning data was drawn exclusively from publicly available, permissively licensed document corpora. Performance figures for initial cross-referencing and QA tasks are design-informed projections grounded in published LoRA scaling benchmarks, while model latency, context ingestion, and generation throughput have been empirically validated on-device. The system is intended to augment, not replace, human judgment in document review.

## Statement on AI Use

The authors used AI writing assistants to improve clarity and LaTeX formatting in preparation of this manuscript.

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## A System Requirements and Installation

MNIME requires Python 3.10+ and has been tested on Windows 10/11 and Ubuntu 22.04. Two installation channels are available:

1. **Pre-built installer:** MNIME\_Setup.exe (Inno Setup wizard) or MNIME\_installer.exe (custom animated PyQt6 installer), available in the installer/ directory.

2. **Source:** Clone the repository, download MNIME-Core-1.5B-Q4\_K\_M.gguf from Hugging Face ([https://huggingface.co/KyleDeanAI/MNIME-Core-1.5B-Q4\\_K\\_M](https://huggingface.co/KyleDeanAI/MNIME-Core-1.5B-Q4_K_M)), place it in models/, then run setup.bat followed by run.bat.

GPU acceleration is auto-detected at runtime via llama-cpp-python’s CUDA backend. NVIDIA GPU layers are set to -1 by default, offloading as many transformer layers as will fit in available VRAM.

## B Key Performance Optimizations

**C-Accelerated Processing.** Merge, extraction, and compression operations use native C-level PyMuPDF routines, achieving up to 50× throughput improvement over pure-Python PDF libraries.

**O(1) Carousel Operations.** Drag-and-drop re-ordering and card removal use surgical layout-index shifts rather than rebuilding the widget tree, giving constant-time performance regardless of file count.

**In-Memory Caching.** Thumbnails and vector SVG icons are rasterized and pre-scaled once per session, eliminating CPU resampling during scroll and hover events.

**Non-Blocking Workers.** All file processing and NLP inference run in dedicated QThread workers. The main thread handles only UI events, guaranteeing frame-rate consistency.

**Dynamic NLP Memory.** The GGUF model and FAISS index are completely deallocated when NLP is toggled off or the application closes, preventing idle memory consumption.